



Salman Bootwala

Lead Data & AI Engineer

12+ years of experience in **Big Data Engineering and Generative AI**, specializing in architecting **agentic data frameworks**, **LLM-integrated pipelines**, and **RAG-based analytics** using **Spark, Kafka, LangChain, and Azure Cloud**.

✉ salmanbw@gmail.com

📍 Dubai, UAE

🐙 github.com/salmanbw

📞 +91-9731194533, +971-507150253

🌐 linkedin.com/in/salman-bootwala

📄 stackoverflow.com/users/3375762/salmanbw

WORK EXPERIENCE

Senior Data & AI Engineer Emirates NBD

08/2023 - Present

Dubai, UAE

Spark Dynamic Data Ingestion & AI Engineering Platform

Tools Used - Spark, Hive, Claude, LLM, VectorDB, RAG

- Spearheaded a GenAI-driven conversion framework using **Claude** models to autonomously translate Informatica BDM mappings into PySpark code.
- **Architected a centralized Claude Prompting Framework** to standardize and orchestrate prompt workflows across data engineering use cases — including code generation, validation, documentation, and testing leveraging advanced **Claude sub-agents, commands, plugins and MCP**.
- **In-house SQL Optimizer** using LLM (Llama3) models to automatically identify, rewrite, and index long-running Oracle SQL queries, reducing execution time by up to 70% through intelligent LLM analysis.
- **POC for an agentic Text-to-SQL** GenAI framework integrating in-house (Llama3 & Qwen) LLM Agents, RAG Agents, Database Agents, and Response Agents with a RAG-powered metadata layer (Redis Vector DB)
- **InHouse DDI (Dynamic Data Ingestion) tool** - To Load data from multiple source systems (Oracle, Mssql, File etc) to multiple target systems.
- Build a **RAG-based ingestion pipeline** that extracts unstructured data from multiple sources, generates embeddings using **Qwen**, and stores them in a **Redis Vector DB**.

Senior Data Engineer G42-Space42

08/2020 - Present

Abu Dhabi, UAE

Dashboard for Social Media Analytics

Tools Used - Java, Python, Microservices, Azure, Neo4j, Gitlab, Docker, Kubernetes

- Building a **data pipeline** for scraping data from social media platforms (twitter,instagram) and processing across multiple data science apps.
- Integrated in-house **NLP, CV, Sentiment Analysis, and Face recognition models** into real-time pipelines using **Python Flask and REST APIs**.
- Building graph database in **Neo4j** and search query database in **Elasticsearch**, along with performance optimization in ES for faster results.
- Developed and Integrated Various **Microservices in Java** using **REST** Calls, asynchronous Processes and scheduling using **Airflow**.
- Orchestrated deployment of **In-house AI microservices** via **GitLab CI/CD** and **Dockerized Kubernetes** clusters with model versioning workflows.
- Configuration and setup of build pipeline in **Gitlab** that includes security and code check - linting, unit test cases, sonarqube etc
- Built **end-to-end data ingestion pipelines** to extract data from **Azure SQL Managed Instance to Azure Synapse** using **ADF and Azure Databricks**
- **Implemented data cleansing, transformation, and enrichment workflows** in **Azure Databricks** leveraging **Delta Lake architecture (Bronze → Silver → Gold)** in **ADLS Gen2**.

EDUCATION

Bachelor of Technology NIT - Bhopal

06/2009 - 05/2013

SKILLS

Spark Spark Streaming GenAI

LLM RAG Agentic AI

Langchain MCP Hadoop

Kafka Airflow SQL NoSql

RabbitMQ CICD Docker

Kubernetes Druid ETL

Azure Azure Data Factory

Azure Databricks Microservices

CERTIFICATIONS

Cloudera Certified Developer for Apache Hadoop (CCDH-410) (09/2015 - Present)

DP-900: Microsoft Azure Data Fundamentals (08/2025 - Present)

AI-900: Microsoft Azure AI Fundamentals (11/2025 - Present)

PROGRAMMING LANG

Java

● ● ● ● ●

SQL

● ● ● ● ●

Python

● ● ● ● ○

WORK EXPERIENCE

Software Engineer III - Big Data

Walmart Labs

10/2018 - 07/2020

Tools Used - Spark, Spark Streaming, Hive, Druid, Kafka

- Experience in writing complex **SQL** queries in **RDBMS, Hive** and **Spark**.
- **Migration of jobs from hive to spark code optimization** :- Optimized a long running hive job by migrating it into spark and further optimized it using **Spark** reducing the full completion time from 12 hours to 1.5 hours.
- Digital acceleration Analytics :- Metrics to analyze customer & visitors trends, sales and transaction numbers for Walmart services like lists, registry, Walmart pay etc using clickstream & transactions data. Created an end to end ETL data pipeline to load and processed **clickstream data** from on-prem clusters to **Druid** using **Spark** and swift storage.
- Realtime data ingestion from **Kafka** to **Druid** :- Build real-time dashboard for businesses to analyze Walmart sales, transactions and customer visits trend using **Kafka** and **Druid**.
- **Kafka to HDFS spark Job** :- Build a pipeline from Kafka to HDFS using spark for data lake.
- **Event Hubs to ADLS Gen2 Streaming Job** – Designed and implemented a near real-time data pipeline from **Azure Event Hubs** to **ADLS Gen2** using **Databricks Structured Streaming** for processing and persisting clickstream data.

Big Data Engineer

British Telecom

04/2017 - 10/2018

Tools Used - Spark, Spark SQL, Hive, Hbase

- **Framework to load data from Hive to Salesforce** : Fetch the data from SalesforceDB using SalesforceAPI, perform transformation in Spark and optimize the code. Similarly cooking up and transforming the data in Hadoop instance and uploading the data back to Salesforce DB. Been part of design for 4 different projects related to Salesforce data transformation.
- **Implementing Delta lake in HDFS**: Using sqoop for importing the data from Mysql and transformation in spark for delta lake and updating the tables in Hive.
- **SCD Implementation**: Using **Spark Dataframes** to implement SCD1 and SCD2 on structured data.
- **SCD using Hbase** : Create hive on **Hbase** integration to implement SCD1 on Hbase on Hadoop data lake.

Consultant - Big Data

Genpact - Headstrong (FINRA)

08/2013 - 04/2017

Tools Used - Java, Hadoop, Hive, AWS S3, AWS Lambda

- Created Custom Map reduce in Java for data cleansing and processing.
- Involved in the analysis, design and implementation of monthly projects.
- Load and transform large sets of structured & semi structured data.
- Build an end to end data pipeline using oracle, sqoop, map reduce and hive.
- Migration of services from on-prem to **AWS cloud using AWS Lambda**
- Use of **AWS S3, Lambda, SQS, SNS and and RDS** for migration of on-prem data to S3.